

BUSINESS REQUIREMENTS DOCUMENT

Employee Attrition Prediction

Predicting Employee Turnover Using Machine Learning

Prepared for: Project Stakeholders

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1. Executive Summary

The Employee Attrition Prediction system is an AI-powered analytics platform designed to help organizations anticipate and reduce employee turnover. By analyzing historical HR data such as employee demographics, tenure, compensation, performance ratings, satisfaction scores, and work patterns, the system identifies employees who are at high risk of leaving and highlights the key factors driving attrition risk. Instead of reacting to resignations after they happen, HR teams and managers can proactively intervene with retention strategies, career development plans, or compensation reviews for employees flagged as high-risk.

Project Objective

Reduce voluntary employee turnover, improve workforce planning, lower recruitment and training costs, and enable data-driven HR decision-making through predictive analytics and explainable AI.

2. Problem Statement

Employee attrition is a significant challenge for organizations of all sizes, resulting in increased recruitment costs, loss of institutional knowledge, reduced team productivity, and disruption to ongoing projects. Traditional HR analysis relies on historical exit interviews and periodic surveys, which are reactive in nature and fail to identify at-risk employees before they decide to leave.

Many organizations lack a systematic, data-driven approach to identify patterns behind attrition, such as low job satisfaction, limited career growth, excessive overtime, compensation gaps, or poor manager relationships. Manual review of HR records across large workforces is time-consuming and often fails to capture the combined effect of multiple contributing factors.

There is a growing need for an intelligent solution that can analyze employee data, predict attrition risk in advance, and explain the underlying reasons, enabling HR teams to take timely, targeted retention action while maintaining data privacy and security.

3. Vision

Our Vision

To build an intelligent, transparent, and reliable AI platform that empowers organizations to understand, predict, and reduce employee attrition, fostering a more stable and engaged workforce.

4. Mission

Our Mission

To provide HR teams and business leaders with a predictive analytics platform that delivers accurate, explainable attrition risk insights instantly, while ensuring data privacy, reliability, and ease of use.

5. Goals

- Optimize workforce retention by identifying at-risk employees before they resign.

- Enhance HR decision-making through AI-driven predictive analytics.
- Deliver data-grounded attrition risk scores based on verified employee records.
- Enable intelligent, explainable identification of the key drivers behind attrition risk.
- Ensure secure and confidential handling of sensitive employee data through robust authentication and data protection mechanisms.
- Provide a scalable, HR-centric analytics platform with an intuitive dashboard for efficient use.

6. Project Objectives

Goals describe the desired business outcomes, while objectives describe the specific actions and capabilities the project will deliver to achieve those goals. Keeping them distinct makes this BRD more professional.

- Develop an AI-powered analytics platform for predicting employee attrition risk.
- Implement secure data ingestion and processing for supported employee data formats (CSV and Excel).
- Enable intelligent risk scoring by leveraging machine learning models (Logistic Regression, Random Forest, XGBoost) to classify attrition likelihood.
- Provide explainability of predictions using SHAP values to improve transparency and trust in AI-generated risk scores.
- Design a responsive and HR-centric dashboard that ensures seamless data management and interaction.

7. Target Users

The system is designed for multiple categories of users, including:

Primary Users	Secondary Users
• HR Managers	• Business Owners / Founders
• HR Business Partners	• Workforce Planning Analysts
• Talent Acquisition Teams	• Compliance Officers
• Department Heads / People Managers	• Data Science Teams
• Chief Human Resources Officers (CHRO)	• Executive Leadership

8. Stakeholders

The success of the project depends on collaboration between multiple stakeholders:

Stakeholder	Role in the Project
Business Owner	Defines business goals and approves scope and budget
Project Manager	Oversees planning, execution, and delivery timelines
AI/ML Development Team	Builds and trains the attrition prediction and explainability models
Frontend Development Team	Develops the HR-facing dashboard application
Backend Development Team	Builds APIs, business logic, and system integrations
UI/UX Designers	Design intuitive and accessible dashboard experiences
Database Administrator	Manages data storage, integrity, and performance
Testing Team (QA)	Validates functionality, performance, and security
End Users (HR Teams)	Use the platform to analyze employees and act on attrition risk insights

9. Proposed Solution

The Employee Attrition Prediction system provides an AI-based analytics dashboard where HR teams can upload employee datasets and receive risk-based insights instantly. Instead of manually reviewing employee records, users can explore questions such as:

- Which employees are at high risk of leaving this quarter?
- What are the top factors driving attrition in the Sales department?
- How does overtime affect attrition risk for junior employees?
- What would improve retention for a specific employee profile?

The AI analyzes the uploaded employee data, computes an attrition risk score for each employee, and generates explanations based solely on the underlying data while highlighting the contributing factors.

10. Scope

In Scope	Out of Scope
• User registration and authentication	• Providing legal or employment law advice
• Upload employee HR datasets (CSV/Excel)	• Automatically terminating or promoting employees
• AI-powered attrition risk prediction	• Predicting individual salary negotiations
• Explainable AI insights (SHAP-based)	• Generating legally binding HR letters
In Scope	Out of Scope

• Employee-level risk dashboard	• Representing the organization in labor disputes
• Department and team-level analytics	
• Secure data storage	
• Responsive web application	
• HR admin dashboard	
• Filter and search employees by risk level	

11. Prioritized Features (MoSCoW Prioritization)

Must Have	Should Have	Could Have	Won't Have (v1)
User Login & Authentication	Attrition Trend Reports	Voice-Based Query Assistant	Automated Employee Termination
Upload CSV/Excel Employee Data	Department-wise Filtering	Multi-language Support	Salary Negotiation Automation
Attrition Risk Prediction	Download Reports	What-if Scenario Simulator	Legal / HR Policy Drafting
Risk Score Dashboard	Multiple Dataset Management	Smart Retention Notifications	Court / Legal Case Prediction
Explainable Risk Factors (SHAP)	Historical Trend Comparison	Chatbot-Style HR Assistant	
Secure Data Storage			

12. Functional Requirements

Data Management

- Upload CSV and Excel employee datasets
- Store uploaded datasets securely
- Delete datasets
- Rename uploaded datasets
- View and preview uploaded datasets

AI Attrition Prediction

- Generate attrition risk score for each employee
- Classify employees into risk categories (Low / Medium / High)
- Display key contributing factors per employee (SHAP explainability)
- Handle batch predictions for multiple employees in one upload
- Inform users if data is insufficient for a reliable prediction

Reporting & Dashboard

- Save prediction history
- View previous prediction runs
- Delete prediction history
- Export reports as PDF or CSV

Search & Filter Functionality

- Search employees by ID, name, or department
- Filter employees by risk level
- Filter datasets by name or upload date

Administration

- Manage users
- Monitor uploaded datasets
- View system activity
- Manage permissions

13. Non-Functional Requirements

Category	Requirements
Performance	Response time less than 5 seconds; fast dataset upload; efficient batch prediction performance

Security	User authentication; data encryption; secure data storage; HTTPS communication; role-based access control
Reliability	High system availability; automatic backup; error recovery mechanisms
Scalability	Support thousands of employee records; handle multiple concurrent dataset uploads; cloud-ready architecture
Category	Requirements
Usability	Clean interface; easy navigation; mobile responsive; simple dashboard interactions
Compatibility	Chrome, Firefox, Microsoft Edge, Safari; desktop and mobile devices
Maintainability	Modular architecture; easy model updates; error logging; documentation

14. Business Rules

- Users must authenticate before uploading employee datasets.
- AI predictions must be based only on the uploaded, verified employee data.
- Employee data remains private and accessible only to authorized users.
- Insufficient or unsupported data should return an informative response rather than a fabricated prediction.
- Uploaded files must comply with supported formats (CSV/Excel).

15. Assumptions

- Users have internet access.
- Uploaded datasets are properly formatted and not corrupted.
- AI models are properly trained on representative historical HR data.
- Users upload employee data they are authorized to access.

16. Constraints

Important

The AI cannot make final HR decisions or provide legal employment advice. Accuracy depends on the quality and completeness of historical HR data, large datasets may require longer processing time, and internet connectivity is required at all times.

17. Technology Stack (Suggested)

Layer	Technologies
Frontend	HTML, CSS, JavaScript, React.js, Bootstrap / Tailwind CSS
Backend	Python (FastAPI or Flask)
AI / ML	Logistic Regression, Random Forest, XGBoost, SHAP for explainability
Database	PostgreSQL / MongoDB
Layer	Technologies
Storage	Cloud Storage (AWS S3 or equivalent)

18. Expected Benefits

For Users	For Organizations
• Faster identification of at-risk employees	• Lower recruitment and training costs
• Data-backed retention conversations	• Increased workforce stability
• Reduced manual review effort	• Enhanced HR compliance
• Improved decision-making	• Improved productivity
• Clear, explainable risk insights	• Secure management of confidential employee data

19. Future Enhancements

- Multi-language dashboard support.
- Voice-based query interaction.
- What-if scenario simulation for retention strategies.
- AI-powered automated retention recommendations.
- Integration with HRIS/payroll systems for real-time data sync.